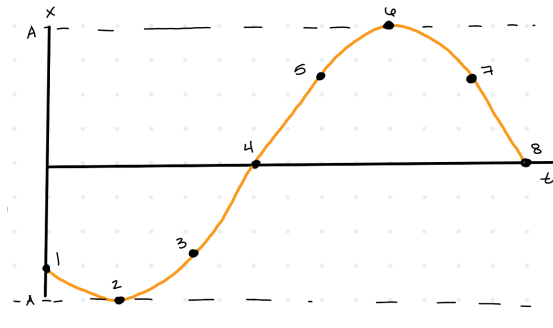


1 Some Quick Conceptual Questions [10 points]

- a (i) $F = -10x$ is in the same format as the equation for the restoring force of a spring using Hooke's Law, $F = -kx$, where $k = 10$ N/m. Said differently, the force is proportional to the negative of the displacement from equilibrium which leads to simple harmonic oscillations.
- (ii.) The time period is calculated using the equation $T = \frac{2\pi}{\omega}$ where $\omega = \sqrt{\frac{k}{m}}$ therefore $2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.1}{10}} = 0.2\pi \approx 0.6$ sec.

Rubric: 1 points for correct force expression for SHM. No points if incorrect. 1 point for computing the time period. 0.5 point taken away if unit is missing. 0.5 point if incorrect equation is used (such as inverting m and k etc).

- b First, we need to understand how acceleration relates to position with SHM. Using our knowledge of kinematics and the motion of a spring we can say: $F = ma = -kx$ if we solve for a we get: $a = \frac{-k}{m}x = -\omega^2x$. Since a and x have a negative relationship, we can envision the x vs t graph as an inverted of the a vs t graph pictured. So to answer (i) the point $x = -A$ is at point 2 and $x = A$ is at point 6. (ii) Remember on x vs t graph, the slope at any given point is the value of v . So at point 8, the slope is negative, therefore the velocity is negative. (iii) At point 3 on our x vs t , the position is between $x = -A$ and $x = 0$.



Rubric: 1 point for each sub-part. 0 for incorrect answer. If justification is incorrect/missing, appropriate points may be taken away (such as 0.5 for each subpart)

- c The out-of-phase mode has the higher frequency. Physically, consider the two normal modes:
- **In-phase mode:** Both pendulums swing together in the same direction at every instant. The distance between the masses does not change, so the coupling spring is neither stretched nor compressed. Therefore, the spring exerts no additional restoring force. The motion is effectively that of two independent simple pendulums, so the frequency equals the natural pendulum frequency.
 - **Out-of-phase mode:** The pendulums swing in opposite directions. In this case, the spring is alternately stretched and compressed, so it provides an additional restoring force on each mass in addition to gravity. This increases the effective restoring force.

Since the angular frequency satisfies

$$\omega = \sqrt{\frac{k_{\text{eff}}}{m}},$$

a larger effective restoring constant implies a larger frequency.

Because the spring contributes only in the out-of-phase motion, that mode has the higher frequency.

Rubric: Full 2 points for correct answer with correct justification. 1 point for correct answer but missing/incorrect justification (may be explained in a way somewhat different than outlined here, such as writing frequency expression)). 0.5 for incorrect answer if something reasonable is attempted.

- d First, let's remember some key relationships: The format of all of these waves is $kx - \omega t$, the wavelength is $\lambda = \frac{2\pi}{k}$, the frequency is $f = \frac{\omega}{2\pi}$ and velocity is $v = \lambda f$. (i) To find which wave correlates with which equation we need to calculate their wavelengths: (a). $\lambda = \frac{2\pi}{2} = \pi$ m (b.) $\lambda = \frac{2\pi}{4} = \frac{\pi}{2}$ m (c.) $\lambda = \frac{2\pi}{8} = \frac{\pi}{4}$ m Therefore, (a) correlates with wave 2, (b.) correlates with wave 3, and (c). correlates with wave 1. (ii) To find their speeds we can do the following: a. $v = \lambda f = \pi \frac{2}{\pi} = 2$ m/s b. $v = \lambda f = \frac{\pi}{2} \frac{4}{\pi} = 2$ m/s c. $v = \lambda f = \frac{\pi}{4} \frac{8}{\pi} = 2$ m/s. As we can see, all of these waves have the same speed.

Rubric: 1.5 points for each part. For part (i.), full points for correct answer with complete justification. 1 point for correct answer without justification. 0.5 point for incorrect answer but something reasonable is attempted. Same for part (ii.)

2 The Power Line [10 Points]

The equation of a catenary is given by

$$y(x) = a \left(\cosh\left(\frac{x}{a}\right) - 1 \right), \quad (1)$$

where $a > 0$ is a constant. A particle of mass m slides without friction along the catenary under the influence of gravity.

- a The gravitational potential energy at a point only depends upon the vertical height from the point of zero gravitational potential energy, so gravitational potential energy of a particle at height y is

$$U(x, y) = mgy, \quad (2)$$

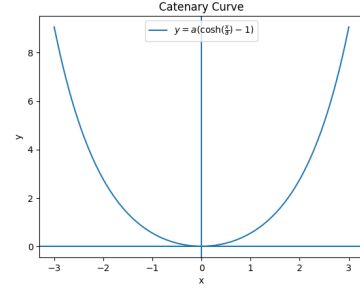
with the reference $U(0, 0) = 0$. Since the particle is constrained to move along the catenary, we substitute

$$y = a \left(\cosh \frac{x}{a} - 1 \right).$$

Thus, the potential energy as a function of x is

$$U(x) = mga \left(\cosh \frac{x}{a} - 1 \right). \quad (3)$$

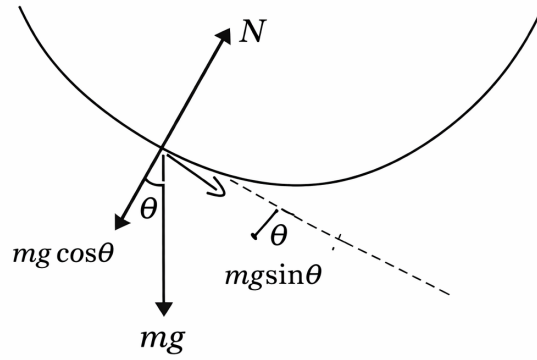
Rubric: 1 point each for writing $U(x, y)$, and for substituting $y(x)$ to get $U(x)$. 0.5 if incorrect answer but something reasonable is attempted.



b At any general point on the catenary ,balancing the force component,

$$N = mg \cos \theta \quad (4)$$

$$ma = mg \sin \theta \quad (5)$$



The equilibrium position occurs at an extremum of the potential energy:

$$\frac{dU}{dx} = 0. \quad (6)$$

$$\frac{d}{dx} [mga (\cosh \frac{x}{a} - 1)] = 0 \quad (7)$$

$$mga \frac{d}{dx} [\frac{(e^{\frac{x}{a}} + e^{-\frac{x}{a}})}{2} - 1] = 0 \quad (8)$$

Differentiating,

$$mg [\frac{(e^{\frac{x}{a}} - e^{-\frac{x}{a}})}{2}] = 0 \quad (9)$$

or,

$$e^{\frac{x}{a}} - e^{-\frac{x}{a}} = 0 \quad (10)$$

$$e^{\frac{2x}{a}} - 1 = 0 \quad (11)$$

$$e^{\frac{2x}{a}} = 1 \quad (12)$$

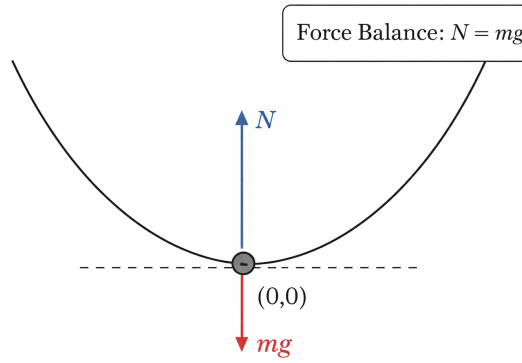
taking log both side, (note $\ln 1 = 0$)

$$\frac{2x}{a} = 0$$

implies that ,

$$x = 0$$

since from eq. (1) at $x = 0$, $y = 0$ Hence, the equilibrium point is



$$(x, y) = (0, 0).$$

As we can see from figure, at this point, $\theta = 0$
so using equations (4),

$$\boxed{N = mg}. \quad (13)$$

i.e. the tangent to the curve is horizontal, and gravity acts vertically downward. Therefore, the normal force balances the weight.

Rubric: 1 point for identifying (0,0) as point of equilibrium. 0.5 point if correct but incorrect/missing justification. 0 point for incorrect answer. 1 point for $N = mg$ condition. 0 point for incorrect answer. No need to do derivatives, a simple force balance is enough as well

- c To study small oscillations, we expand the potential energy about $x = 0$. Using the Taylor expansions
since,

$$\begin{aligned} \cosh z &= \frac{e^z + e^{-z}}{2} \\ \cosh z &= \frac{(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots) + (1 - z + \frac{z^2}{2!} - \frac{z^3}{3!} + \dots)}{2}, \end{aligned}$$

we obtain

$$\cosh z = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} \dots$$

so,

$$\cosh\left(\frac{x}{a}\right) = 1 + \frac{x^2}{2a^2} + \frac{x^4}{24a^2} + \dots$$

for small oscillations (x is very small), so neglecting higher order terms,

$$\cosh\left(\frac{x}{a}\right) - 1 \approx \frac{x^2}{2a^2} \quad (14)$$

Substituting into $U(x)$,

$$U(x) \approx mg a \cdot \frac{x^2}{2a^2} = \frac{mg}{2a} x^2. \quad (15)$$

Thus, near equilibrium,

$$\boxed{U(x) \approx \frac{1}{2} \left(\frac{mg}{a} \right) x^2}. \quad (16)$$

This is the standard quadratic form of a simple harmonic oscillator potential, confirming simple harmonic motion for small deviations.

Rubric: 2 points for correct Taylor expansion. One point for identifying linear term to vanish due to equilibrium. 1 point for putting everything together to get quadratic potential. 0.5 point to be taken away for each minor mistake. If Taylor expansion is incorrect, but idea is correct, 1.5 points to be taken off.

d Comparing expression of potential energy with the SHM potential

$$U(x) = \frac{1}{2} k x^2,$$

we identify

$$k = \frac{mg}{a}.$$

Hence, the angular frequency of oscillation is

$$\boxed{\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{a}}}. \quad (17)$$

Rubric: Full 2 points for correct answer with proper matching with the standard SHM condition/derivation/argument. 1 point to be taken away if the factor of 1/2 is incorrectly accounted for since this impacts Problem 3. 0.5 if incorrect answer but something reasonable if attempted.

3 The Power Line, Again – With Friction [10 points]

a The catenary is given by

$$y(x) = a \left(\cosh \left(\frac{x}{a} \right) - 1 \right)$$

For small oscillations about the lowest point $x = 0$, we assume $|x| \ll a$.

(i) Friction force effectively points along the x -direction

The slope of the catenary is

$$\frac{dy}{dx} = \sinh \left(\frac{x}{a} \right)$$

For small x/a ,

$$\sinh \left(\frac{x}{a} \right) \approx \frac{x}{a}$$

Thus, the tangent angle θ satisfies

$$\tan \theta \approx \frac{x}{a}$$

For small oscillations, $\theta \ll 1$, so

$$\cos \theta \approx 1, \quad \sin \theta \approx \frac{x}{a}$$

Since friction acts along the tangent direction, its horizontal component is

$$F_{f,x} = F_f \cos \theta \approx F_f$$

The vertical component is of higher order in x/a and can be neglected. Hence, to leading order, the friction force may be taken along the x -direction.

(ii)

The forces acting on the particle are:

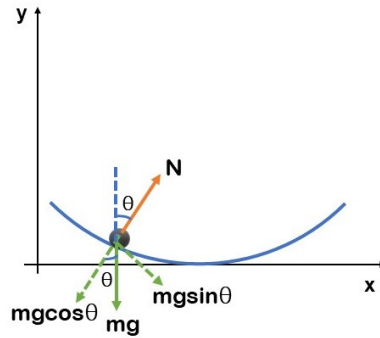
- Gravitational force mg (downward),
- Normal force N (perpendicular to the curve).

For small oscillations, the wire is nearly horizontal, so θ is small. Balancing forces along the radial direction where the particle does not move:

$$N = mg \cos \theta$$

Since $\cos \theta \approx 1$,

$$N \approx mg$$



Thus, to leading order,

$$N \approx mg$$

(iii) **Effective spring constant**

Expand the catenary for small x/a :

$$\cosh\left(\frac{x}{a}\right) \approx 1 + \frac{1}{2}\left(\frac{x}{a}\right)^2$$

Substituting into $y(x)$:

$$y(x) \approx a \left(1 + \frac{x^2}{2a^2} - 1 \right) = \frac{x^2}{2a}$$

Thus, the height above equilibrium is

$$y \approx \frac{x^2}{2a}$$

The gravitational potential energy is

$$U = mgy = mg \frac{x^2}{2a}$$

This has the form

$$U = \frac{1}{2} k x^2,$$

from which we identify

$$\boxed{k = \frac{mg}{a}}$$

Rubric: 1 point for each subpart. Small angle approximation, and showing $\cos \theta \approx 1$ and appropriate components suffices. For part (iii.) students may refer to problem 2. No extra points for incorrect k be taken away if mistake in problem 2 but the work following is consistent. 0.5 marks may be taken away for each minor error or incomplete justification where needed. 0.5 marks total if all incorrect but something reasonable is attempted.

- b For small oscillations about $x = 0$, the motion reduces to an effective one-dimensional oscillator along the x -direction with

$$k = \frac{mg}{a}, \quad N \approx mg.$$

The forces along x are:

- Restoring force: $F_s = -kx = -\frac{mg}{a}x$
- Kinetic friction: $F_f = -\mu_k mg \operatorname{sgn}(\dot{x})$

Using Newton's second law,

$$m\ddot{x} = -\frac{mg}{a}x - \mu_k mg \operatorname{sgn}(\dot{x})$$

Dividing by m ,

$$\boxed{\ddot{x} + \frac{g}{a}x + \mu_k g \operatorname{sgn}(\dot{x}) = 0}$$

Since friction changes sign depending on the velocity, we write two separate equations:

Case 1: $\dot{x} > 0$ (motion to the right)

$$m\ddot{x} = -\frac{mg}{a}x - \mu_k mg$$

or

$$\ddot{x} + \frac{g}{a}x + \mu_k g = 0.$$

Case 2: $\dot{x} < 0$ (motion to the left)

$$m\ddot{x} = -\frac{mg}{a}x + \mu_k mg$$

or

$$\ddot{x} + \frac{g}{a}x - \mu_k g = 0$$

Rubric: 1 point each for having the two conditional equations. 1 point will be taken away if conditions are mixed up/similar error. Substitution for k explicitly not required. 0.5 marks total if overall incorrect but something reasonable is attempted.

c Minimum displacement required for motion

If the particle begins at rest, static friction acts. Motion starts only when the restoring force exceeds the maximum static friction force.

The restoring force magnitude is

$$|F_s| = k|x| = \frac{mg}{a}|x|.$$

The maximum static friction force is

$$F_{s,\max} = \mu_s N$$

Since for small oscillations $N \approx mg$,

$$F_{s,\max} = \mu_s mg$$

For motion to begin,

$$\frac{mg}{a}|x| > \mu_s mg$$

Canceling mg ,

$$\frac{|x|}{a} > \mu_s$$

Thus,

$$|x| > a\mu_s$$

$$x_{0(\min)} = a\mu_s$$

Therefore, the particle must be displaced by at least $a\mu_s$ from equilibrium in order to overcome static friction and start moving.

Rubric: Full 1 point for correct answer with basic force balance justification. Quoting AJP paper with the D condition also acceptable. 0.5 point is correct but no justification. 0 points if incorrect.

d For small oscillations, the motion reduces to

$$m\ddot{x} = -\frac{mg}{a}x - \mu_k mg \operatorname{sgn}(\dot{x}),$$

with

$$k = \frac{mg}{a}, \quad \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{a}}$$

Between two successive turning points x_0 and x_1 , the speed is zero at both points. Using energy conservation, including work done by friction:

$$\frac{1}{2}kx_1^2 - \frac{1}{2}kx_0^2 = \mu_k mg |x_1 - x_0|.$$

If the particle crosses the origin, then x_0 and x_1 have opposite sign

$$|x_1 - x_0| = |x_0| + |x_1|.$$

Thus,

$$\frac{1}{2}kx_1^2 - \frac{1}{2}kx_0^2 = \mu_k mg(|x_0| + |x_1|).$$

Using $k = \frac{mg}{a}$, substitute into the equation:

$$\frac{1}{2} \frac{mg}{a} x_0^2 - \frac{1}{2} \frac{mg}{a} x_1^2 = \mu_k mg(x_0 + x_1).$$

Divide both sides by mg :

$$\frac{x_0^2}{2a} - \frac{x_1^2}{2a} = \mu_k(x_0 + x_1).$$

Multiply throughout by $2a$:

$$x_0^2 - x_1^2 = 2a\mu_k(x_0 + x_1).$$

Factor the left-hand side:

$$(x_0 - x_1)(x_0 + x_1) = 2a\mu_k(x_0 + x_1).$$

Cancel $(x_0 + x_1)$ from both sides:

$$x_0 - x_1 = 2a\mu_k.$$

Hence,

$$\boxed{x_1 = x_0 - 2\mu_k a.}$$

Condition to cross the origin

Thus, for an oscillator with constant-magnitude friction, the amplitude decreases by a fixed amount during each half-cycle: $2\mu_k a$

say,

$$C = 2a\mu_k.$$

Therefore, after the first half-cycle, the new turning point is

$$x_1 = x_0 - C.$$

For the particle to cross the origin at least once, we must have

$$x_1 > 0.$$

Thus,

$$x_0 - 2a\mu_k > 0$$

$$\boxed{x_0 > 2a\mu_k}$$

Time to stop for the first time

Within each half-cycle, the motion is simple harmonic with angular frequency

$$\omega = \sqrt{\frac{g}{a}}$$

The time between successive turning points is

$$t_{\text{half}} = \frac{\pi}{\omega}.$$

Hence,

$$\boxed{t_1 = \frac{\pi}{\omega} = \pi \sqrt{\frac{a}{g}}}$$

This time is independent of the friction coefficient.

Rubric: 1 point for each part. Full 1 points for correct answer with proper justification from first principle. Quoting from AJP paper allowed. 0.5 point for correct answer without justification/proper citation. No points if incorrect.

e Given:

$$a = 500 \text{ m}, \quad \mu_s = 0.5, \quad \mu_k = 0.4, \quad m = 1 \text{ kg}, \quad g \approx 10 \text{ m/s}^2, \quad x_0 = 450 \text{ m}.$$

Effective spring constant

$$k = \frac{mg}{a} = \frac{1 \times 10}{500} = 0.02 \text{ N/m}.$$

Angular frequency:

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{a}} = \sqrt{\frac{10}{500}} = \sqrt{\frac{1}{50}} = \frac{1}{\sqrt{50}}.$$

Using $\sqrt{50} \approx 7$,

$$\omega \approx \frac{1}{7}$$

Amplitude decrease per half-cycle

For constant-magnitude friction,

$$C = \frac{2\mu_k mg}{k}.$$

Since $k = \frac{mg}{a}$,

$$C = 2a\mu_k$$

$$C = 2(500)(0.4) = 400 \text{ m}$$

Thus, the amplitude reduces by 400 m per half-cycle.

Number of origin crossings

$$\frac{x_0}{C} = \frac{450}{400} = 1.125$$

The largest integer less than this is 1.

Therefore, the particle crosses the origin once.

After one half-cycle,

$$x_1 = x_0 - C = 450 - 400 = 50 \text{ m}$$

Stopping condition

Minimum displacement needed to overcome static friction:

$$x_{\min} = a\mu_s = 500(0.5) = 250 \text{ m}$$

Since

$$50 < 250,$$

The restoring force cannot overcome static friction, So the particle stops there.

Time to complete stop

Each half-cycle takes

$$t_{\text{half}} = \frac{\pi}{\omega} = \pi \sqrt{\frac{a}{g}}.$$

$$\sqrt{\frac{a}{g}} = \sqrt{\frac{500}{10}} = \sqrt{50} \approx 7.$$

Using $\pi \approx 3$,

$$t_{\text{half}} \approx 3 \times 7 = 21 \text{ s.}$$

Since motion stops after one half-cycle,

$$t_{\text{stop}} \approx 21 \text{ s.}$$

Rubric: 1 point each. 0.5 point taken away for minor numerical mistakes/lack of units. Full points for correct answer with proper justification from first principle. Quoting from AJP paper allowed. No point for each part if part is incorrect.

4 The Power Line, Yet Again – With a Tuned Mass Damper [10 points]

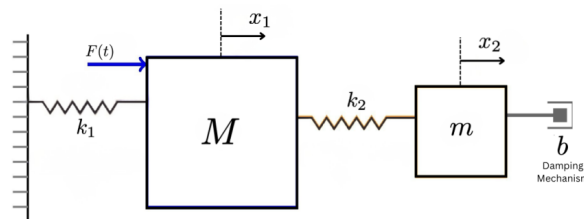


Figure 1: Schematic of the system

- a Working in the complex plane, we can write the force of the wind on the power line as the real part of

$$F(t) = F_0 e^{i\omega t} \quad (18)$$

There are three forces acting on the primary mass M:

- Natural restoring force $-k_1 x_1$

- External force (wind) $F(t)$
- Coupling force $-k_2(x_1 - x_2)$

So, the equation of motion of primary mass M can be written as:

$$\boxed{M\ddot{x}_1 = -k_1x_1 - k_2(x_1 - x_2) + F_0e^{i\omega t}} \quad (19)$$

There are two forces acting on the secondary mass m :

- Coupling force $-k_2(x_2 - x_1)$
- Viscous damping $-b\dot{x}_2$.

Now, the equation of motion for secondary mass m can be written as,

$$\boxed{m\ddot{x}_2 = -k_2(x_2 - x_1) - b\dot{x}_2} \quad (20)$$

Rubric: 1.5 points for equation of motion of each mass. 0.5 marks to be taken away for each minor mistake in either of the equations, eg. sign error, missing a term, etc. Detailed justifications for signs, etc. not explicitly required.

- b The steady state solution will contain only the driving frequency of the wind, ω . The can be written as,

$$x_1 = A_1e^{i\omega t}, \text{ and } x_2 = A_2e^{i\omega t} \quad (21)$$

Where A_1 and A_2 are complex amplitudes.

Now, compute the derivatives with respect to time:

$$\dot{x}_j = i\omega A_j e^{i\omega t} \implies \ddot{x}_j = -\omega^2 A_j e^{i\omega t} \quad (22)$$

.

Let's now substitute this in the equations of motion from part (a) and cancel out the exponential factor which can't be zero.

First, the equation for the primary mass M :

$$M(-\omega^2 A_1 e^{i\omega t}) + (k_1 + k_2)A_1 e^{i\omega t} - k_2 A_2 e^{i\omega t} = F_0 e^{i\omega t} \quad (23)$$

$$\implies (k_1 + k_2 - M\omega^2)A_1 - k_2 A_2 = F_0 \quad (24)$$

Next, for the secondary mass m ,

$$m(-\omega^2 A_2 e^{i\omega t}) + b(i\omega A_2 e^{i\omega t}) + k_2 A_1 e^{i\omega t} = 0 \quad (25)$$

$$\implies -k_2 A_1 + (k_2 - m\omega^2 + ib\omega)A_2 = 0 \quad (26)$$

Now, if we want to write this in matrix form

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix} \quad (27)$$

We get,

$$\begin{pmatrix} k_1 + k_2 - M\omega^2 & -k_2 \\ -k_2 & k_2 - m\omega^2 + ib\omega \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} F_0 \\ 0 \end{pmatrix} \quad (28)$$

And comparing the coefficients we get,

$$a = k_1 + k_2 - M\omega^2, \quad (29)$$

$$b = -k_2 \quad (30)$$

$$c = -k_2, \quad (31)$$

$$d = k_2 - m\omega^2 + ib\omega, \quad (32)$$

$$e = F_0, \quad (33)$$

$$f = 0 \quad (34)$$

Rubric: 1 point for correct substitution and simplification to get linear equations in A_1 and A_2 . 1 point for writing as matrix and identifying the terms. Masses can be divided out as well. 0.5 point in each step for minor mistakes. 0.5 total if completely incorrect but something is reasonable attempted.

- c Firstly, we ignore the damping in this part. So set $b = 0$. Next, we are operating in the regime where the wind force frequency matches with the resonant frequency of M, that is,

$$\omega \approx \omega_0 \equiv \sqrt{k/M} \quad (35)$$

Under these conditions, the equations of motion become,

$$(k_1 + k_2 - M\omega_0^2)A_1 - k_2A_2 = F_0 \quad (36)$$

$$-k_2A_1 + (k_2 - m\omega_0^2)A_2 = 0 \quad (37)$$

Now, we want the amplitude of the primary to be 0:

$$-k_2A_2 = F_0 \quad (38)$$

$$(k_2 - m\omega_0^2)A_2 = 0 \quad (39)$$

Using the second equation, and observing that something *has to* move, that is, $A_2 \neq 0$, we get:

$$\boxed{k_2 = m\omega_0^2 = mk_1/M} \quad (40)$$

The physical meaning is that the secondary mass must also be attuned to have the same resonant frequency $\sqrt{k_2/m} = \omega_0$ which is able to take out energy from M and oscillate itself.

Rubric: Full 2 points for correct answer with full consistent algebraic steps. 1 point if correct answer but steps and consistency cannot be traced. 0.5 for incorrect answer but something reasonable is attempted.

- d Now, we keep k_2 such that the tuning condition is still maintained, but we account for a damping $b \neq 0$.

Taking the equation of motion for the secondary mass,

$$-k_2 A_1 + (k_2 - m\omega_0^2 + ib\omega_0) A_2 = 0 \quad (41)$$

With the tuning condition $k_2 = m\omega_0^2$,

$$-k_2 A_1 + ib\omega_0 A_2 = 0 \implies \frac{A_1}{A_2} = \frac{ib\omega_0}{k_2} \quad (42)$$

Which is simply,

$$\boxed{\frac{A_1}{A_2} = \frac{ib}{m\omega_0}} \quad (43)$$

So, if damping $b \longrightarrow 0$, the earlier condition of $A_1 \longrightarrow 0$ is recovered.

Rubric: Full 2 points for correct answer with full consistent algebraic steps. 0.5 point if correct answer but steps and consistency cannot be traced. 0.5 for incorrect answer but something reasonable is attempted.

- e Recall that b, m, ω_0 are all real numbers but A_1 and A_2 are complex. Writing $A_1 = C_1 e^{i\phi_1}$ and $A_2 = C_2 e^{i\phi_2}$, we can now write,

$$\frac{C_1}{C_2} = \frac{b}{m\omega_0} \quad (44)$$

and

$$e^{i(\phi_1 - \phi_2)} = i = e^{i\pi/2} \implies \phi_1 - \phi_2 = \pi/2 \quad (45)$$

So, the oscillations of the two masses are out of phase by $\pi/2$.

Rubric: Full 1 point for correct answer with justification of phase difference. 0.5 point if something is on the right track but not fully justified/shown. No points for incorrect answer.
